

Computational Systems Biology for Digital Medicine

7–12 December 2025

Wellcome Genome Campus, UK

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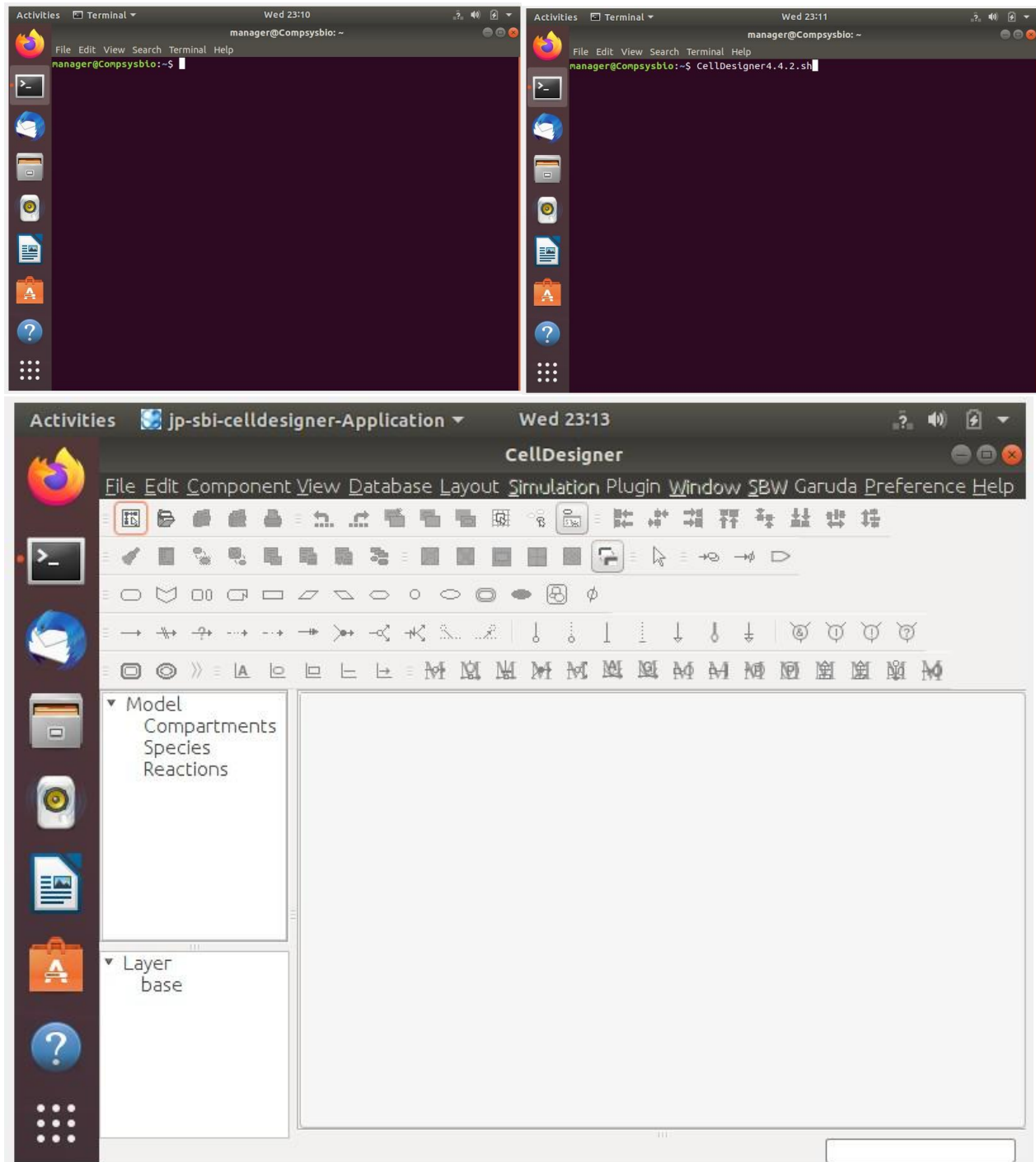
Hands-on session with CellDesigner

In this session, we will see examples of CellDesigner diagram functionalities. We will also open and observe a file containing a graphical model of apoptosis.

The functionalities we will see are summarised below:

Open a new model	Create a reaction	Add anchor point	Add catalytic reaction	Set active state
Change color	Include compartment	Add residue to protein	Change position of residue	Create complex
	Include genes and RNAs	Connect to databases	Choose layout	

Open the terminal in your VM session and type CellDesigner4.4.2.sh or CellDesigner



You will see in your screen the general view as discussed in the demo:

General view

TREE AREA

displays all the list of the components in a tree structure.

DRAW AREA

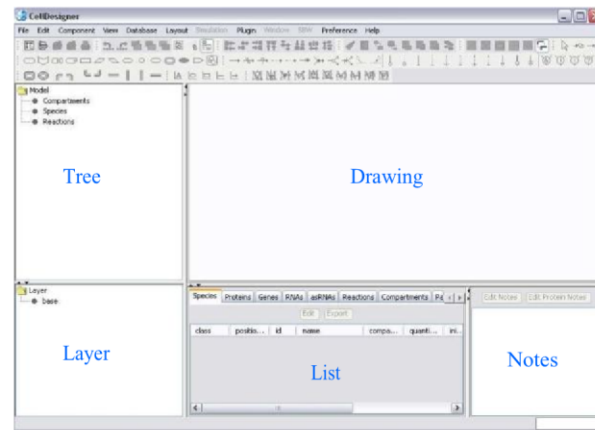
is the area where the model is built from the tool bar items.

LIST AREA

displays and edits the list of the components of the model (species, proteins, genes, RNAs, etc.)

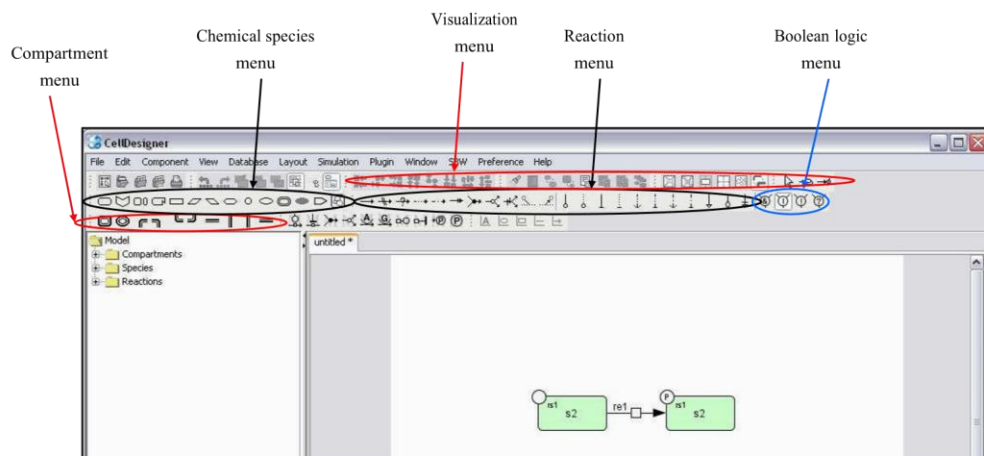
NOTES AREA

displays and edits the notes of each component (reactions, protein, complex, etc.)

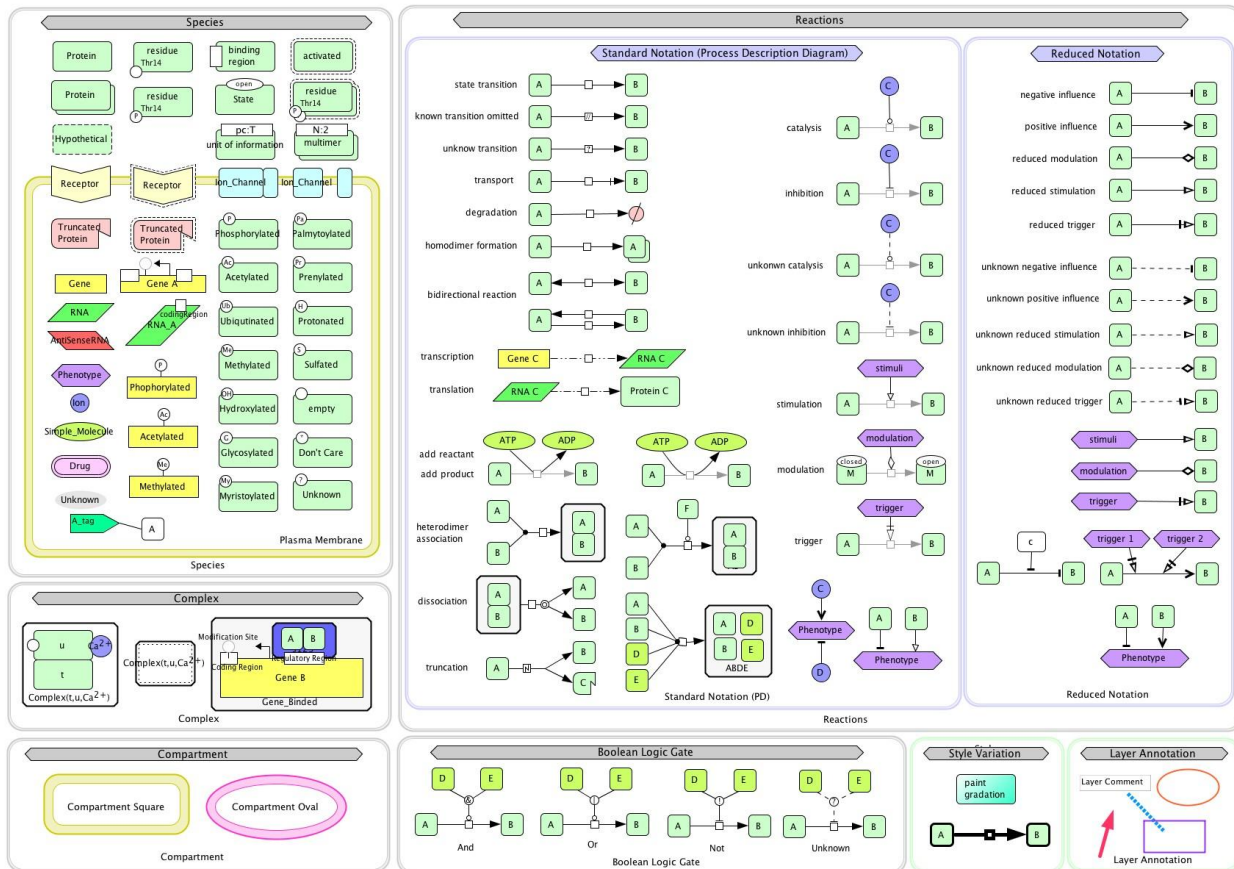


Take some time to explore the menu. Check the glyphs, the arcs and the macros. Highlighted below the Compartment menu, the chemical species menu, the visualization menu, the reaction menu and the Boolean gates menu.

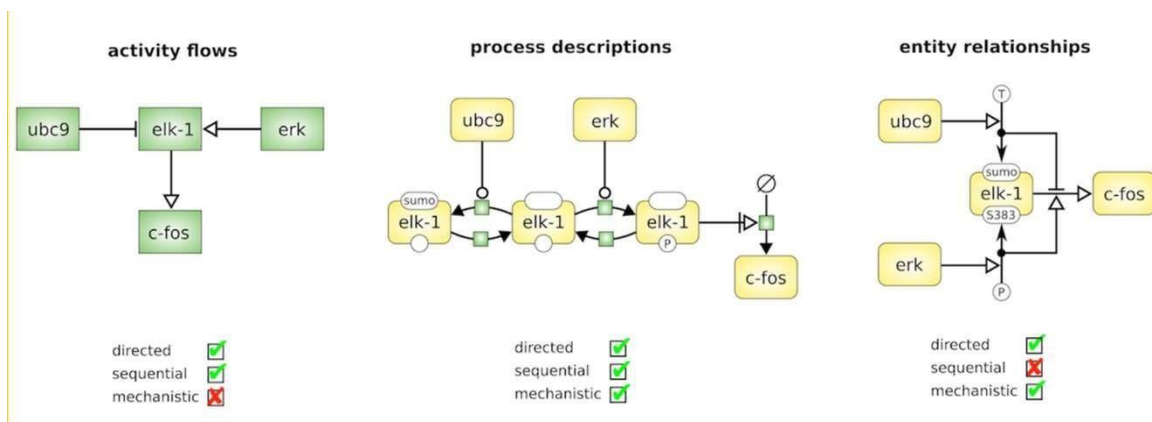
The menu



Here is a summary of all schemas that can be used to create your diagram:
The software suite can accommodate different SBGN schemes such as Process Description or Activity Flow diagrams.



Can you make some comments about the reduced notation scheme? Here are the three different languages of SBGN to help you understand better.



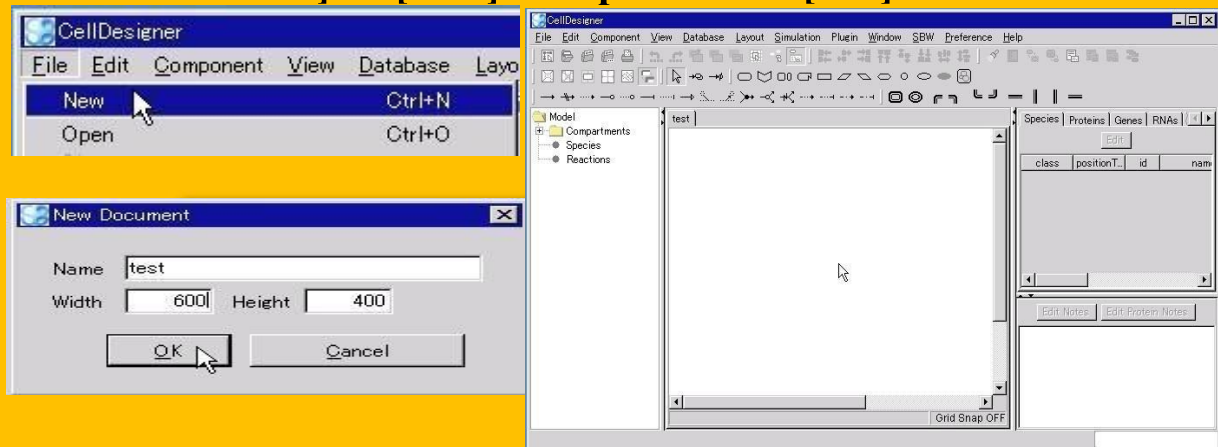
Let's start creating a diagram!

Click on File, then New, then write the title: test and click OK. We will use the default size settings.

Demonstration

● Create new model:

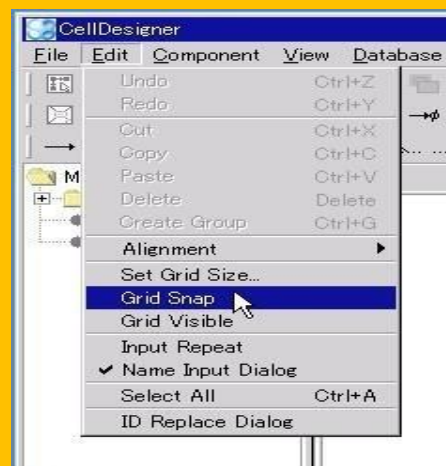
● File] → [New] → input title → [OK]



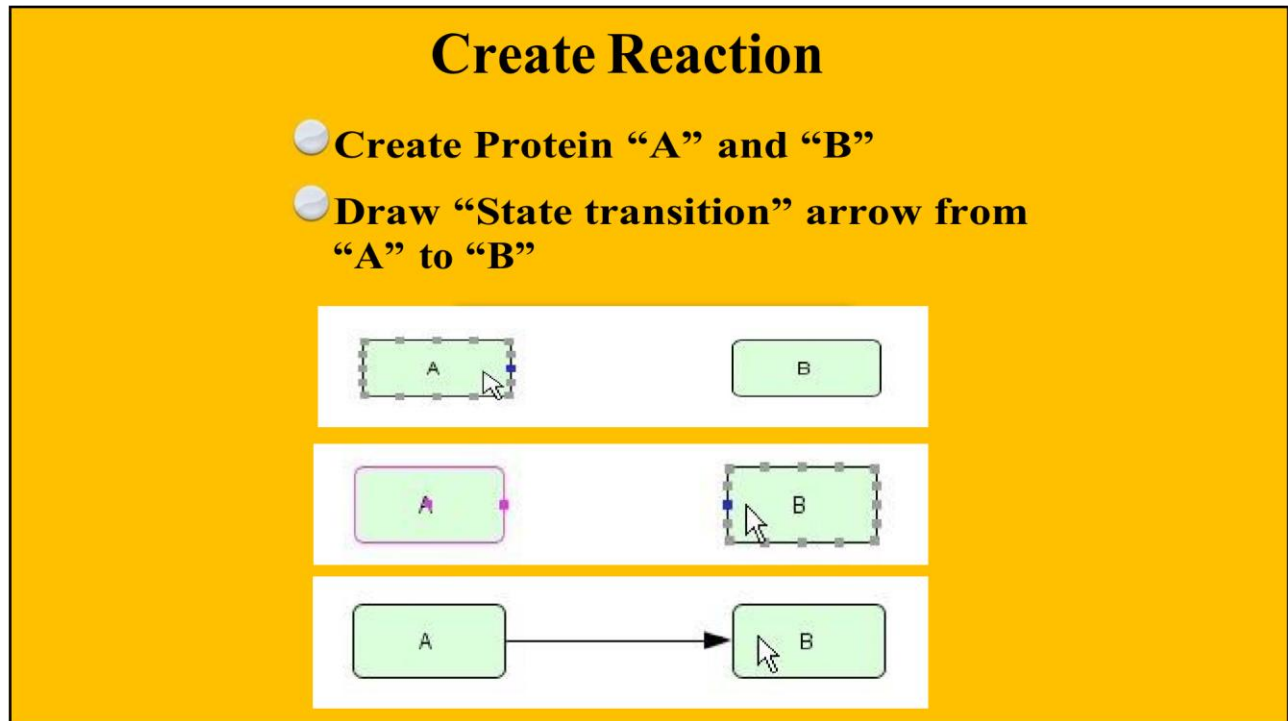
We will also enable the grid to help us draw more aligned diagrams. To do so, we will click on Edit in the Menu bar and then on Grid Snap as seen in the screenshot below:

Tips

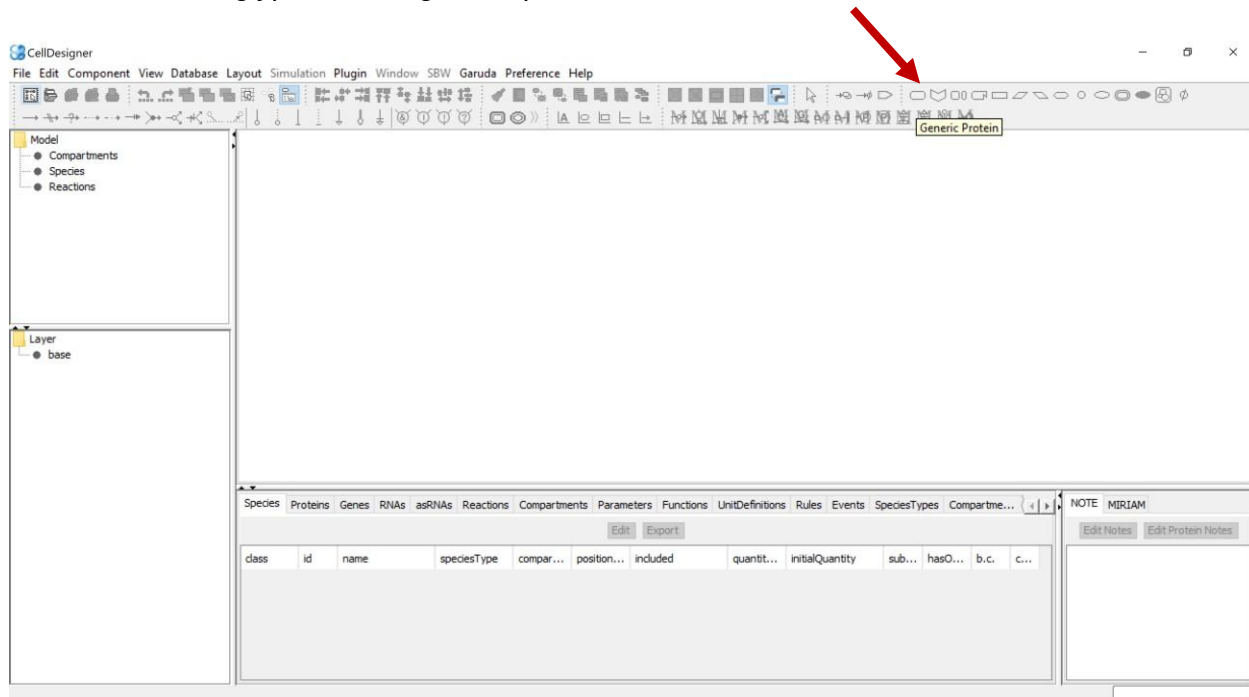
● Enable [Grid Snap] will help you draw your model much easier



Now we are ready to create our first reaction!



Choose from the glyph menu a generic protein :



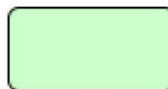
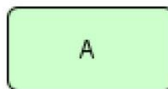
and then click on the window of the drawing space. Name the protein A.



Name of the species ×



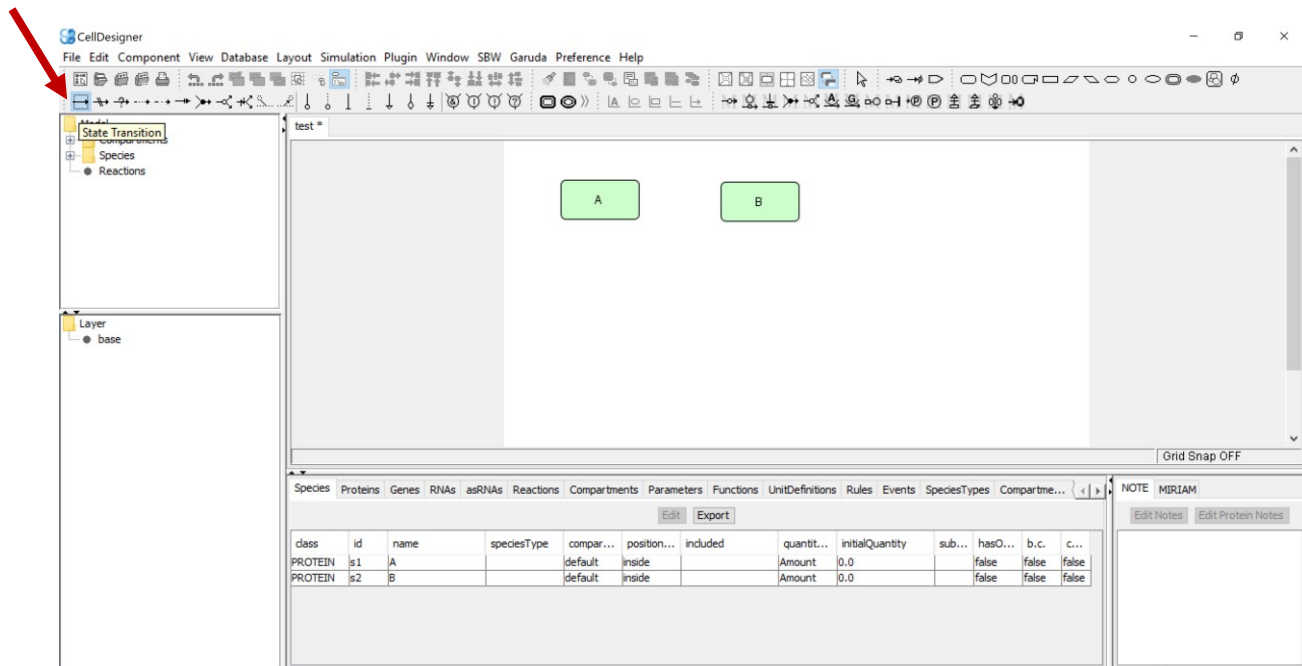
Repeat and name the protein B:



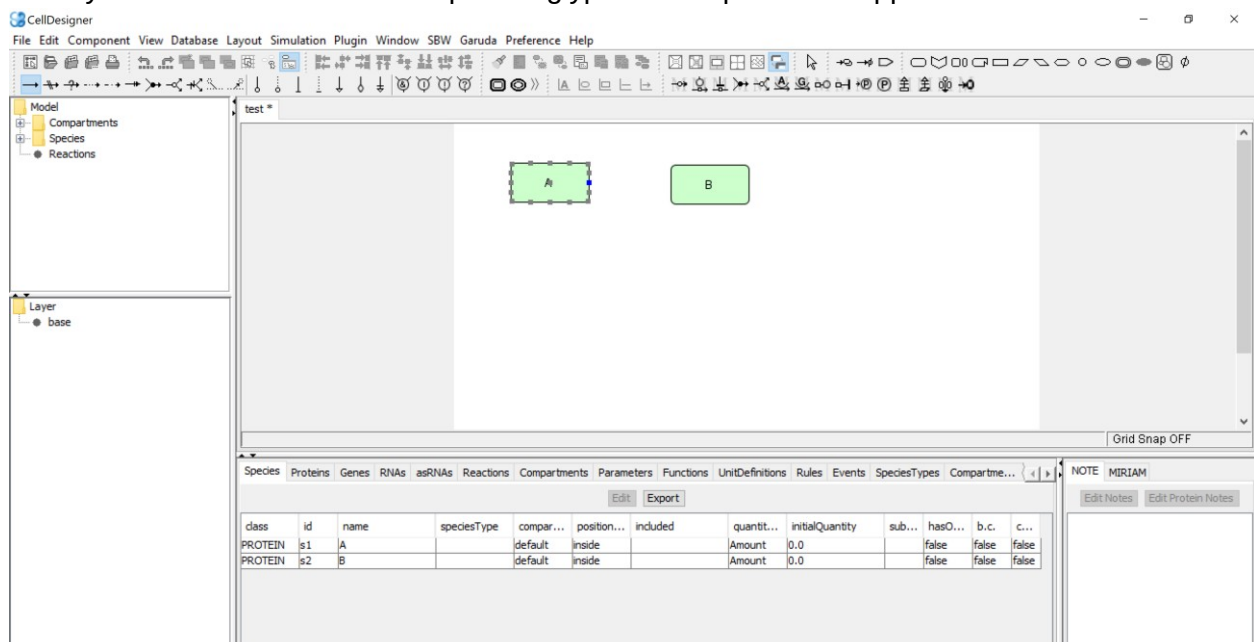
Name of the species ×



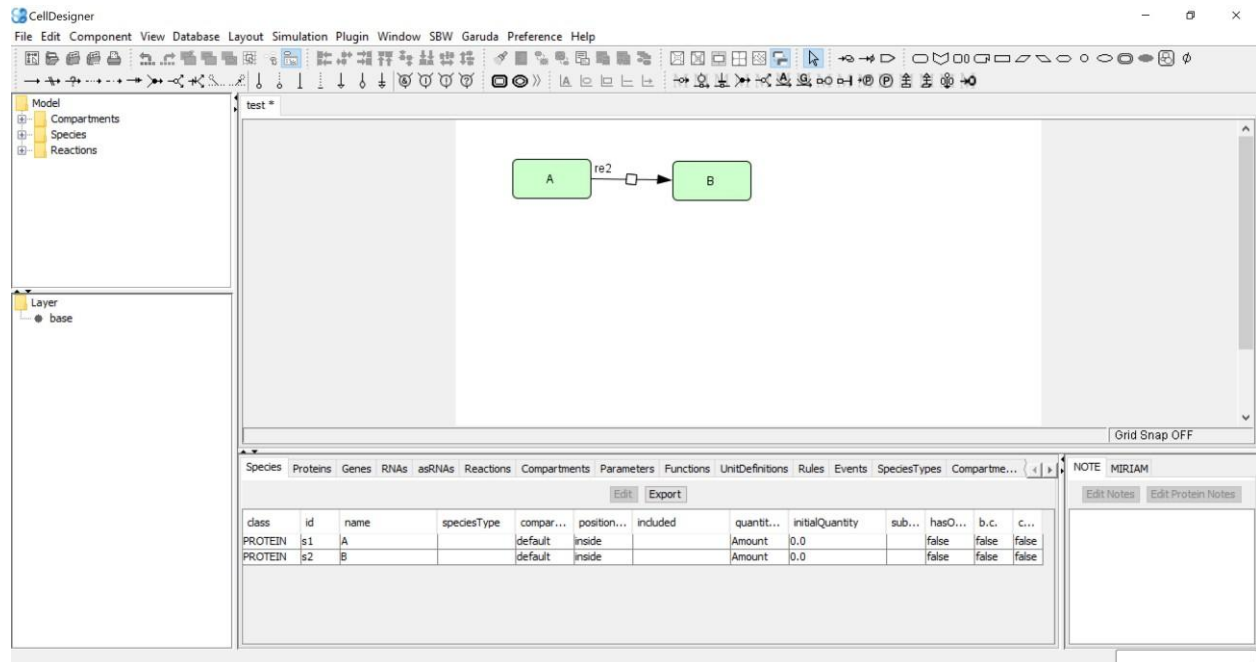
Then click on the state transition arrow



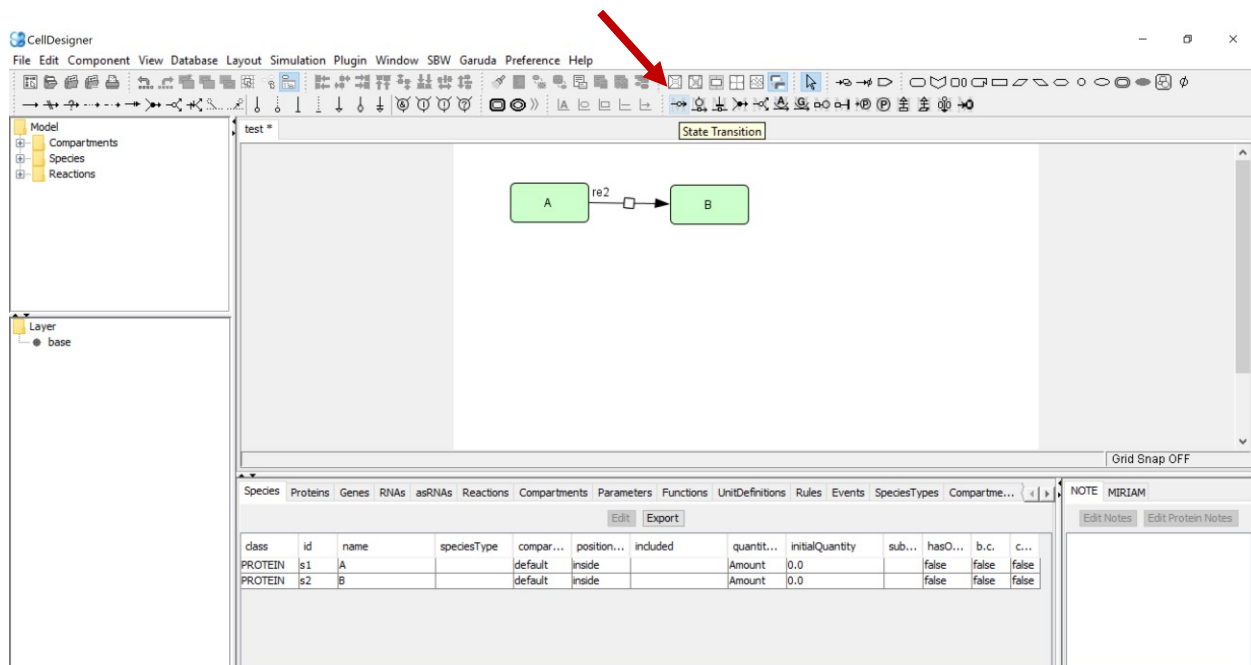
When you move the cursor on the protein glyph anchor points will appear.



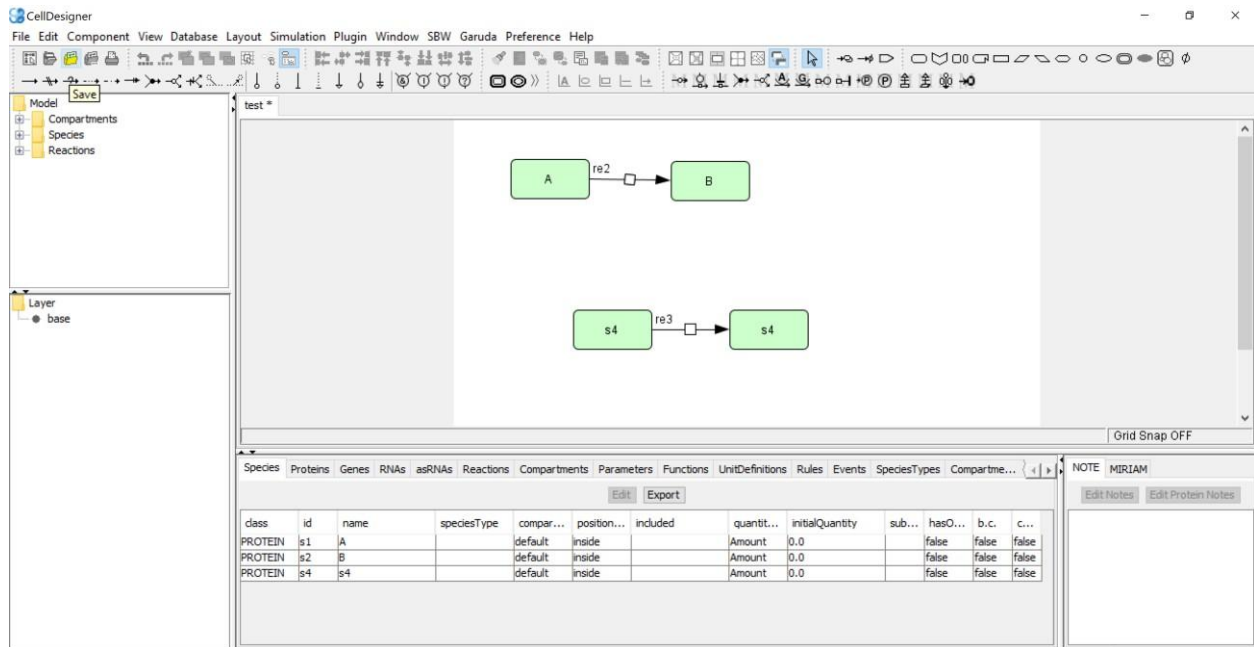
Choose the middle on the right on protein A, click and then choose the middle on the left of protein B and click again to have your arrow drawn.



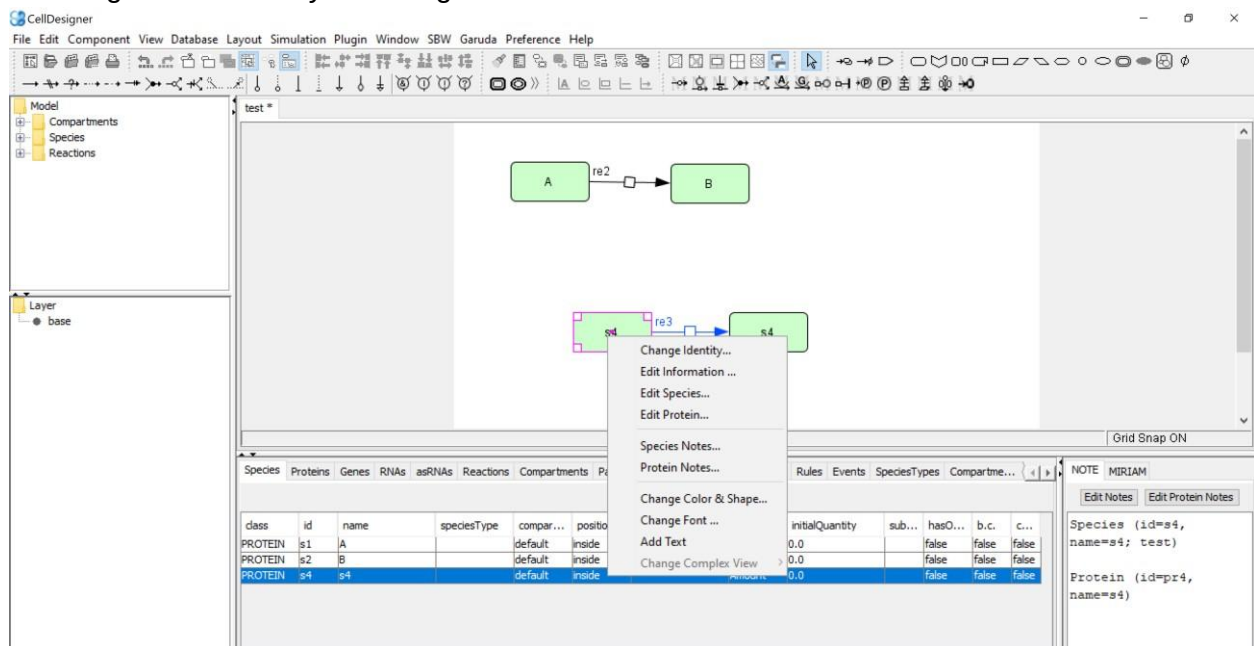
Notice that you can use macros to draw this reaction automatically. Choose the state transition from the macros menu and then click on the drawing space just below the reaction you created:



A state transition reaction is now created automatically:



To change their names you can right click on them and choose from the menu Edit Protein



Now you can replace the s4 generic name with C or whatever name you want. You can name it IL6.

CellDesigner

File Edit Component View Database Layout Simulation Plugin Window SBW Garuda Preference Help

Model

- Compartments
- Species
- Reactions

Layer

- base

test *

Protein

name: s4

type: GENERIC

residues/regions:

add... edit... del...

edit block diagram...

Update Cancel

A --re2--> B

s4 --re3--> s4

Grid Snap ON

Species Proteins Genes RNAs asRNAs Reactions Compartments Parameters Functions UnitDefinitions Rules Events SpeciesTypes Compartme...

Edit Export

class	id	name	speciesType	compar...	positon...	included	quantit...	initialQuantity	sub...	hasO...	b.c.	c...
PROTEIN	s1	A		default	inside		Amount	0.0		false	false	false
PROTEIN	s2	B		default	inside		Amount	0.0		false	false	false
PROTEIN	s4	s4		default	inside		Amount	0.0		false	false	false

NOTE MIRIAM

Edit Notes Edit Protein Notes

Species (id=s4, name=s4; test)

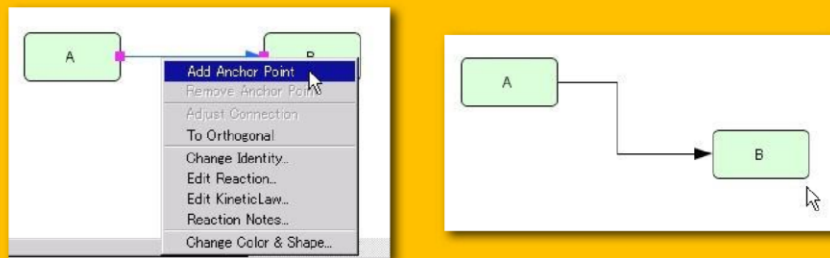
Protein (id=pr4, name=s4)

What do you observe?

We can also add anchor points:

Add Anchor Point

- Add 2 anchor points to reaction
- Drag reaction and anchor point to change its shape



Now let us try to add a protein that has a positive effect (catalyzes) the reaction we created previously. We would need to select a generic protein, click on the drawing space, and then change it's name. The next step would be to select the catalysis arrow and add it:

CellDesigner

File Edit Component View Database Layout Simulation Plugin Window SBW Garuda Preference Help

Model

- Compartment
- Species
- Reactions

Layer

- base

tes Catalysis

class	id	name	speciesType	compar...	position...	included	quantit...	initialQuantity	sub...	hasO...	b.c...	c...
PROTEIN	s1	A		default	inside		Amount	0.0		false	false	false
PROTEIN	s2	B		default	inside		Amount	0.0		false	false	false
PROTEIN	s5	C		default	inside		Amount	0.0		false	false	false

Grid Snap ON

Species Proteins Genes RNAs asRNAs Reactions Compartments Parameters Functions UnitDefinitions Rules Events SpeciesTypes Compartme...

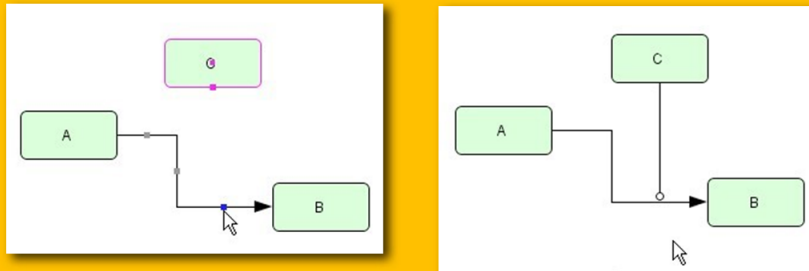
Edit Export

NOTE MIRIAM

Edit Notes Edit Protein Notes

Add Catalysis reaction

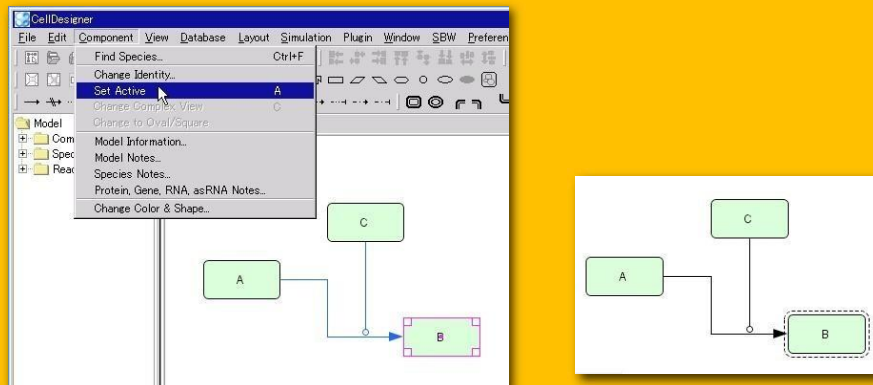
- Add Protein “C”
- Add Catalysis reaction from “C” to the reaction



In CellDesigner you can use dots to indicate an active state, however, this notation is not used in SBGN language

Set Active state

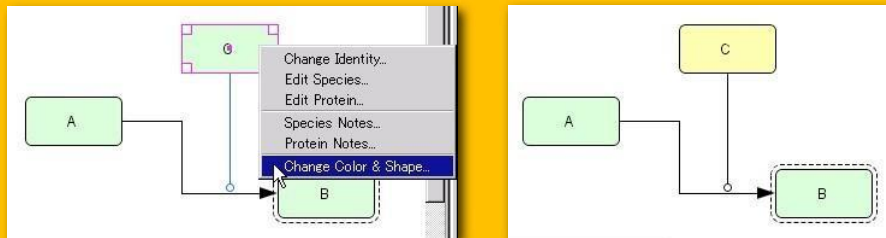
- Select Protein “B”
- [Component] → [SetActive]



You can also change the color by right clicking on the element.

Change Color

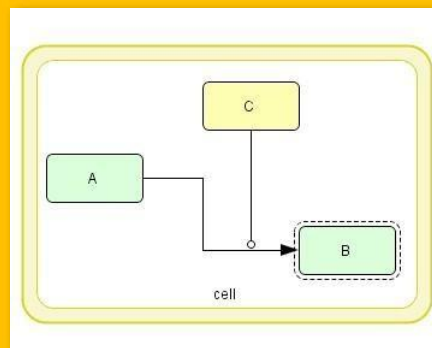
- Right click on Protein “C”
- Select [Change Color & Shape...]



Adding compartments is also important:

Compartment

- Click [Compartment] icon 
- Drag mouse cursor to specify its area
- Input name of compartment

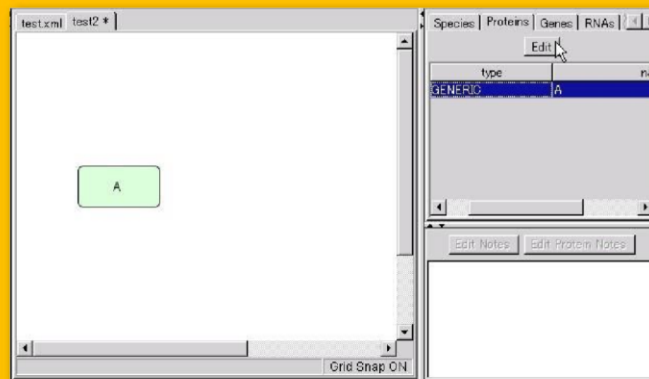


Practice makes perfect !

The following exercises in pages 15 – 18 are for practice at home – let's go to page 19 and connect to databases!

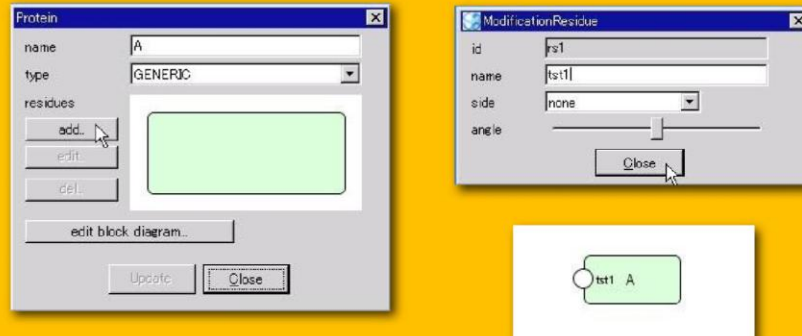
Add Residue to Protein

- Create new model (test2)
- Create Protein “A”
- Select Protein “A” in [Proteins] Tab
- Click [Edit] button



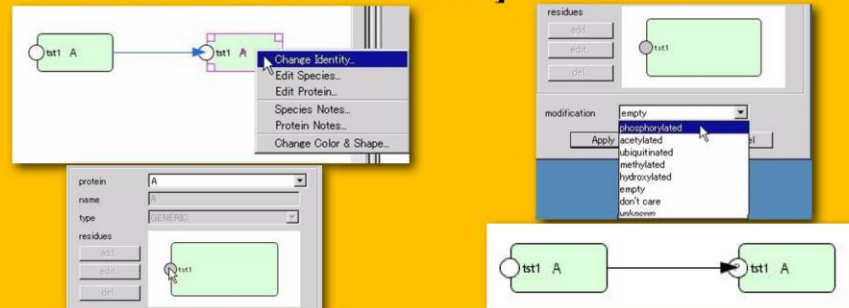
Add Residue to Protein

- Click [add] button on [Protein] dialog
- Input name for the residue (tst1)
- Click [Close] button
- Click [Update] Button



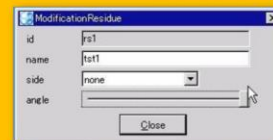
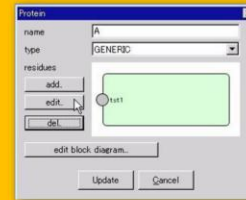
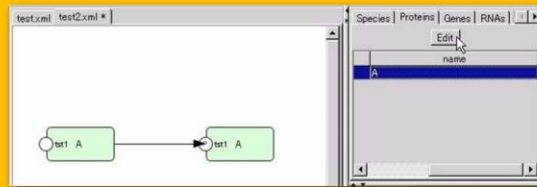
Add Residue to Protein

- Copy & Paste Protein “A” and then draw “State Transition” arrow
- Right Click on “A” (right side) and select [Change Identity..].
- Click residue “tst1” in Dialog
- Select [phosphorylated] in modification



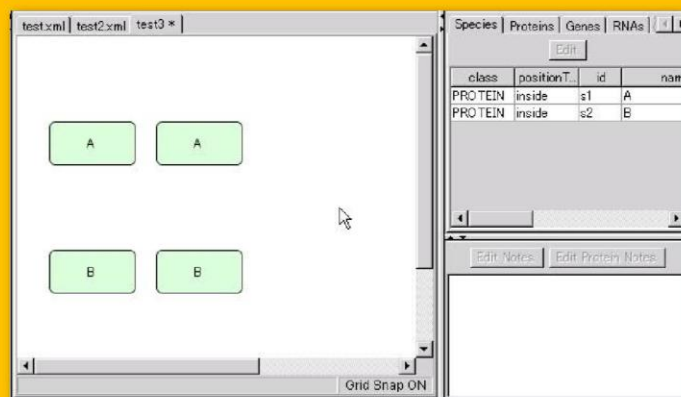
Change position of Residue

- Select Protein “A” in [Proteins] Tab
- Click [Edit] button
- Click residue “tst1” in Dialog
- Click [edit] button
- Drag [angle] sidebar



Complex

- Create new model (test3)
- Create Proteins “A” and “B”
- Copy & Paste both “A” and “B”

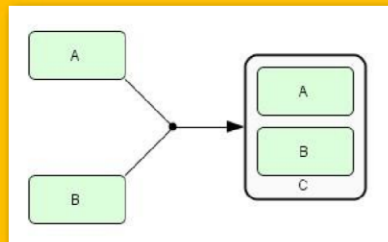


Complex

- Click [Complex] icon and create complex “C”

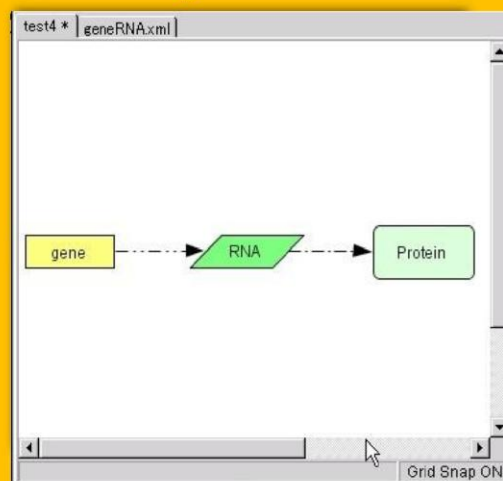


- Drag Protein “A” and “B” into complex C
- Draw “Association” arrow



Gene & RNA

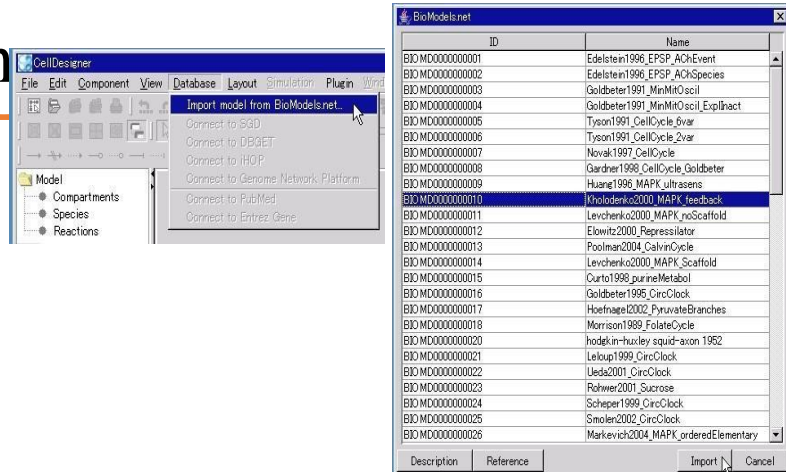
- Create new model (test4)
- Create gene, RNA and Protein
- Draw “Transcription” and “Translation”



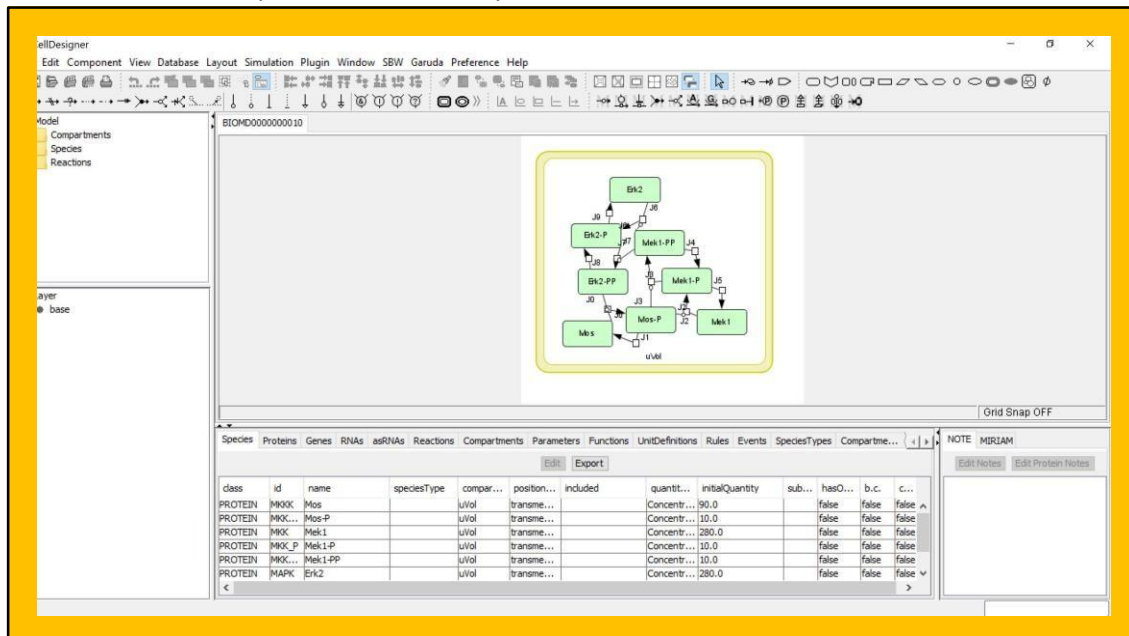
Welcome! Let's see how we can import a model from a database! Start from a clean slate! Create a new model (File, New etc). Then click on the Database tab of the Menu and select a model from BioModels. Here we have selected the MAPK model of Kholodenko (Ultrasensitivity).

Database Connection

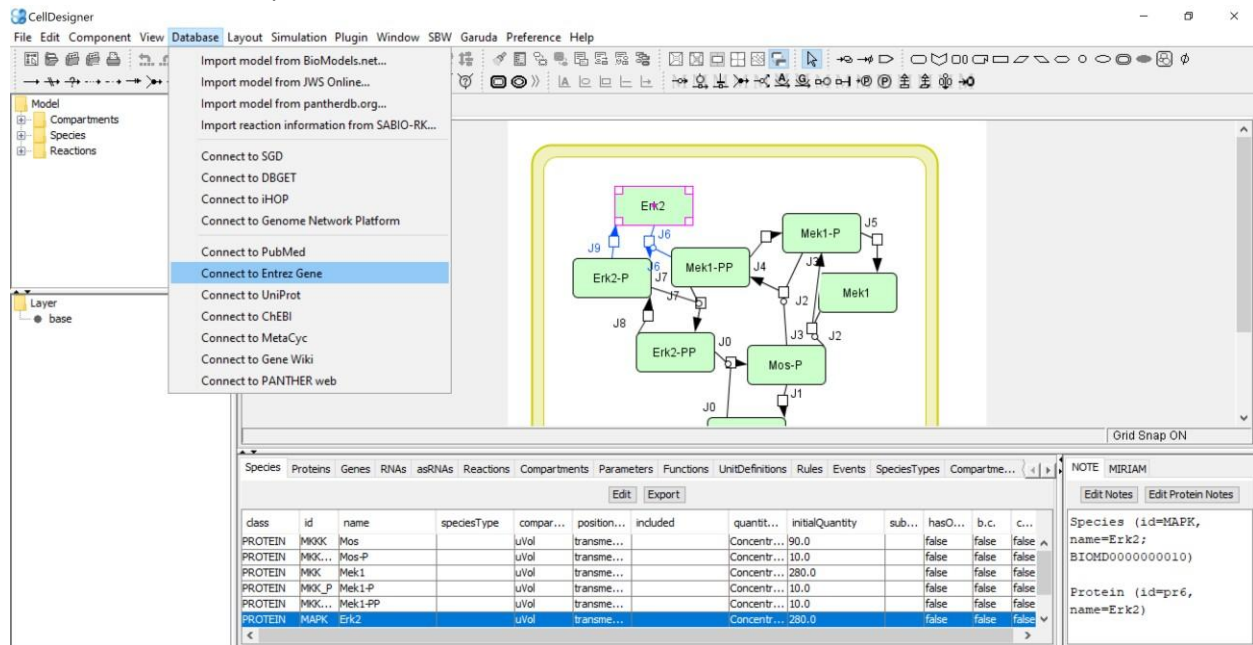
Import model from BioModels.net



The model is then imported automatically! Take some time to observe it!



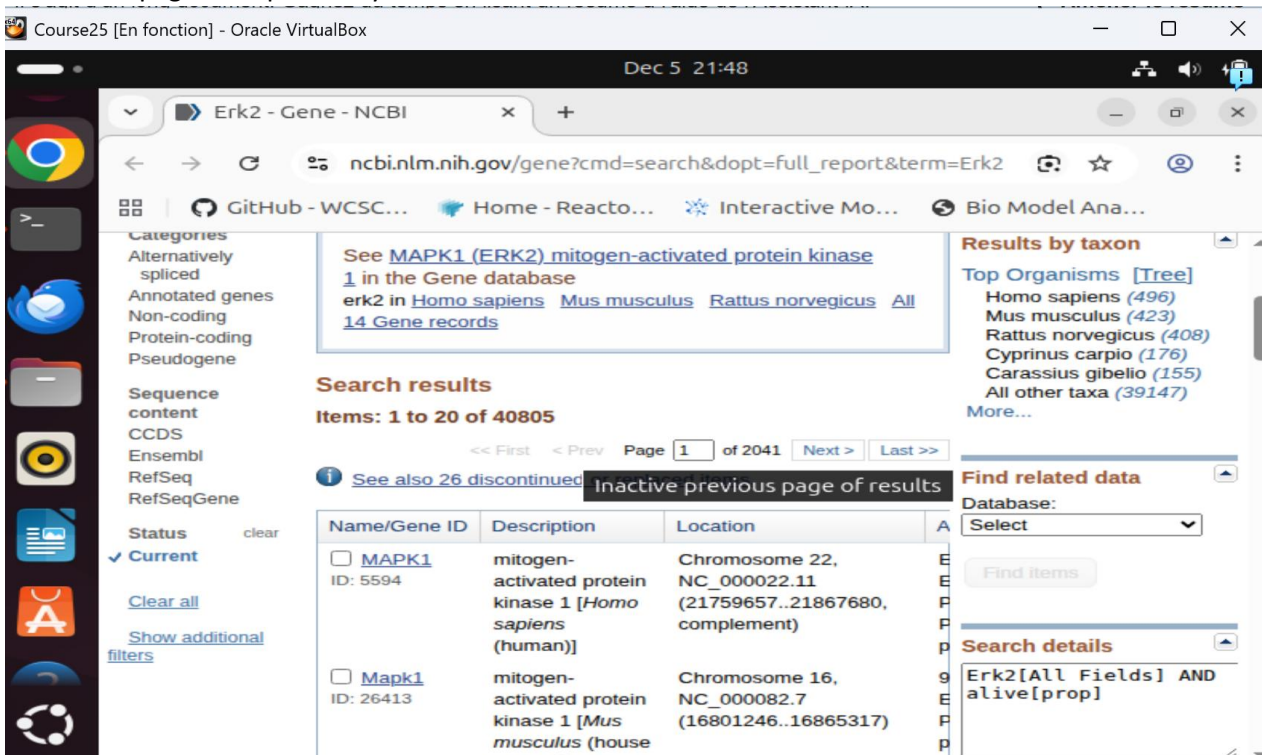
Now click on the ERK2 protein of the model, then click on Databases and select Connect to Entrez Gene



The screenshot shows the CellDesigner interface. The 'Database' menu is open, and 'Connect to Entrez Gene' is highlighted. The main window displays a signaling pathway model with various proteins and their interactions. The 'Species' table at the bottom lists the proteins and their associated data.

class	id	name	speciesType	compar...	position...	included	quantit...	initialQuantity	sub...	hasO...	b.c.	c...
PROTEIN	MOCK	Mos		uVol	transme...		Concentr...	90.0		false	false	false
PROTEIN	MOCK	Mos-P		uVol	transme...		Concentr...	10.0		false	false	false
PROTEIN	MOCK	Mek1		uVol	transme...		Concentr...	280.0		false	false	false
PROTEIN	MOCK	Mek1-P		uVol	transme...		Concentr...	10.0		false	false	false
PROTEIN	MOCK	Mek1-PP		uVol	transme...		Concentr...	10.0		false	false	false
PROTEIN	MAPK	Erk2		uVol	transme...		Concentr...	280.0		false	false	false

A browser page will open and you will be transferred to the Entrez Gene database site:



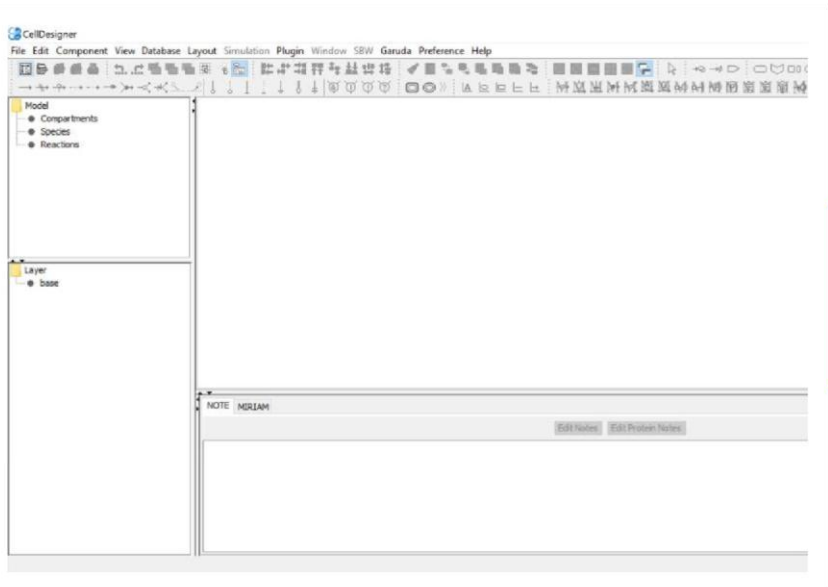
The screenshot shows a web browser window displaying the Entrez Gene database page for ERK2. The page includes search results, a list of organisms, and a table of gene records.

Search results
Items: 1 to 20 of 40805

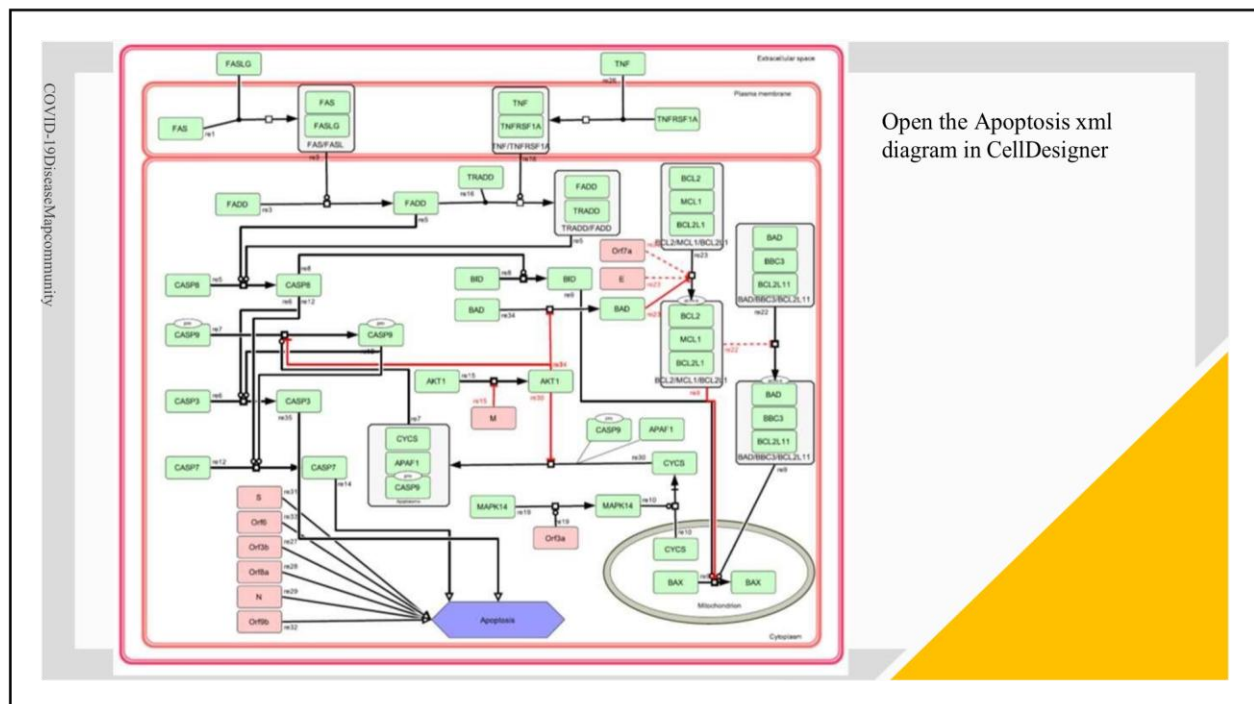
Results by taxon
Top Organisms [Tree]
Homo sapiens (496)
Mus musculus (423)
Rattus norvegicus (408)
Cyprinus carpio (176)
Carassius gibelio (155)
All other taxa (39147)
More...

Name/Gene ID	Description	Location
☐ MAPK1 ID: 5594	mitogen-activated protein kinase 1 [<i>Homo sapiens</i> (human)]	Chromosome 22, NC_000022.11 (21759657..21867680, complement)
☐ Mapk1 ID: 26413	mitogen-activated protein kinase 1 [<i>Mus musculus</i> (house)]	Chromosome 16, NC_000082.7 (16801246..16865317)

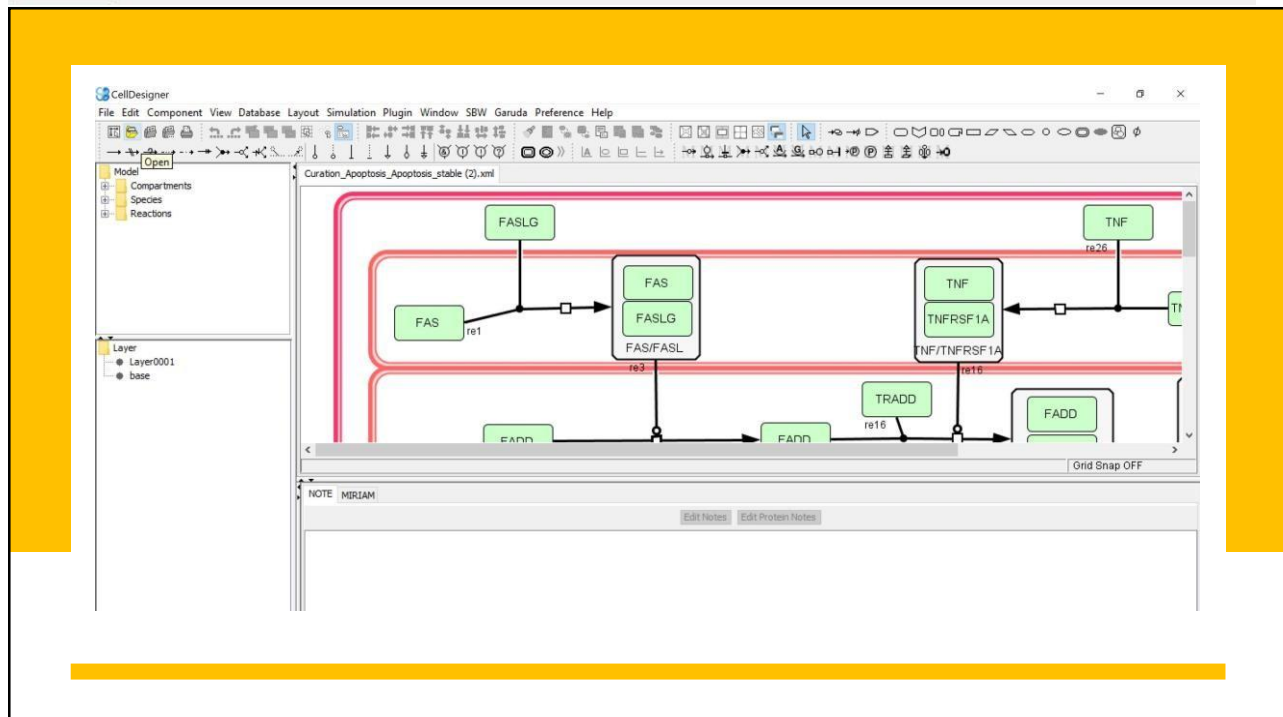
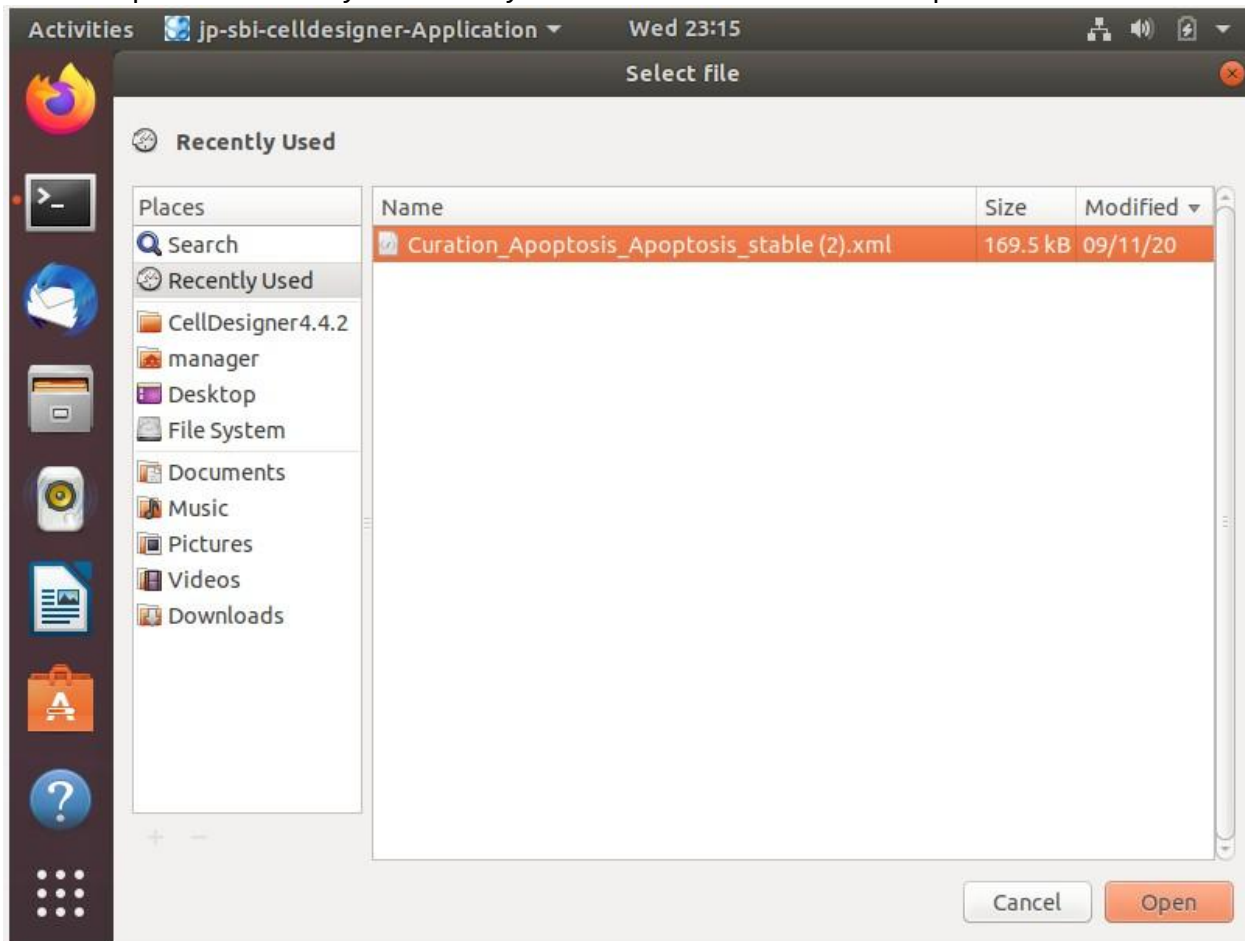
Bonus! Open in CellDesigner the xml file of the Apoptosis module form the COVID19 Disease Map repository. Take some time to observe the layout, the compartments, the species, the annotations. Click on elements and then connect to Entrez Gene or Pubmed to find relevant information!



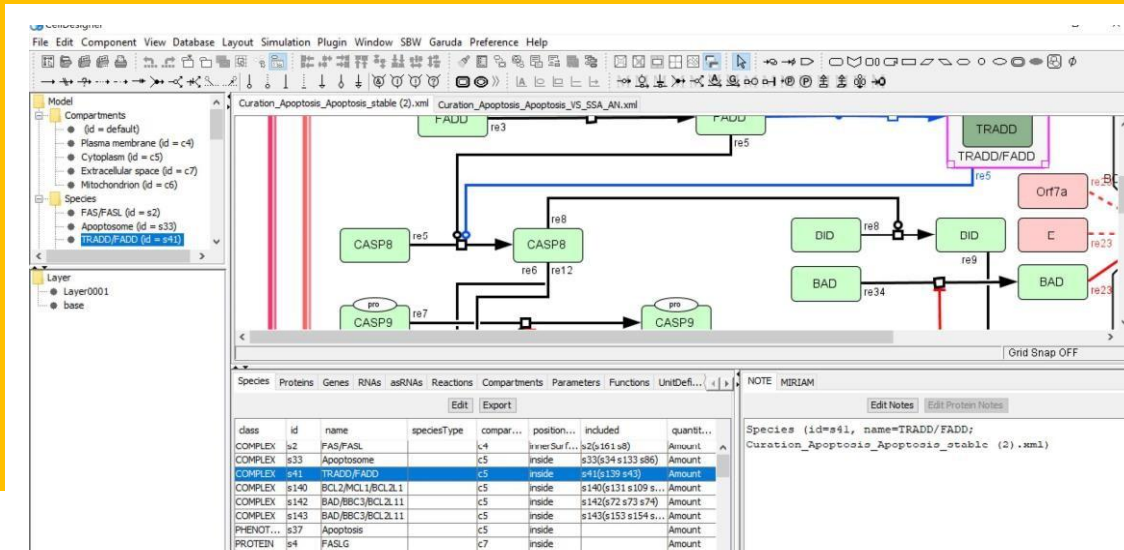
Open in
CellDesigner



File-> Open and search your directory to find the file! Here is an example:



Click on a complex to see the information in the species tab



Click on a reaction to see the MIRIAM annotations!

